

# PIYUSH MUDGAL

AI/ML Engineer | Open Source Engineer | Linux (Arch / Ubuntu)

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## SUMMARY

- **Open Source Contributor** at **Apache Airflow** (44k+ GitHub stars, used by 70,000+ companies) with **25+ PRs merged** across core, providers, and testing, including race-condition fixes and SQLAlchemy hook standardization. Actively working toward **committer status**.
- **AI/ML Engineer** with hands-on experience in **Generative AI, LLM integrations**, and **deep learning**, building production-grade developer tooling and AI-powered applications with cloud deployment and CI/CD exposure.
- **International Research Experience**: ML Research Intern at **Kyushu Institute of Technology**, Japan, co-authoring a paper accepted at **ICIEV/IVPR** on CNN+LSTM-based robotic perception, achieving **95% object detection** and **90%+ gesture recognition** accuracy.

## OPEN SOURCE CONTRIBUTIONS

### Apache Airflow | Contributor

2025 - Present

*Data orchestration platform with 44k+ GitHub stars, used by 70,000+ companies.*

- Fixed a [race condition in FabAuthManager](#) and resolved a startup race condition in the API server, eliminating database integrity errors.
- Implemented SQLAlchemy-compatible `sqaalchemy_url` properties for Presto, Vertica, and Impala hooks, and fixed [Connection.get\\_uri\(\)](#) for non-string extras.
- 25+ PRs merged across core, providers, and testing; engaged with core maintainers to align designs with community norms before opening PRs.

### Observal (BlazeUp-AI) | Contributor | Moderator

2026 - Present

*Self-hosted agent registry and analytics platform for AI coding agents (2.1k+ GitHub stars).*

- Authored unit test suites for `observal_cli`, covering command parsing, output formatting, and edge-case handling.
- Implemented a coverage threshold ratchet to enforce incremental coverage gains and prevent regressions in untested modules.
- Resolved patch-target and mocking issues flagged during PR review, aligning test mocks with `unittest.mock` best practices and project conventions.

## EXPERIENCE

### Compyle (YC-backed) | Software Engineer, Product & DX | Remote

Nov 2025 – Feb 2026

*Next.js, TypeScript, Drizzle ORM, REST APIs, Resend, Vercel, Github Actions, CI/CD*

- Designed and built the Compyle Apps Showcase from scratch, a self-serve publishing platform removing the team's main submission bottleneck.
- Built a reusable component library and email template system with live preview, improving frontend consistency across the platform.
- Authored official engineering documentation covering setup, deployment, and Git practices

### Kyushu Institute of Technology | ML Research Intern | Fukuoka, Japan

Feb – May 2025

*Python, PyTorch, OpenCV, CNN, LSTM*

- Designed and implemented a CNN + LSTM-based robotic perception system, achieving 95% object detection and 90%+ gesture recognition accuracy on a dataset of over 100K samples.
- Redesigned the data preprocessing pipeline, reducing model training time by 60% and enabling significantly faster experimentation cycles.
- Co-authored a research paper accepted at ICIEV/IVPR and presented findings on robotic perception at the conference.

## EDUCATION

### B.Tech, Computer Science & Engineering (AI/ML Specialization)

Nov 2021 - Jul 2025

Dronacharya College of Engineering, Gurugram

CGPA: 7.2 – Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Neural Networks, Operating Systems

## COMMUNITY AND LEADERSHIP

### Observal (BlazeUp-AI) | Moderator & Contributor

2026 – Present

- Review and merge community contributions, write unit tests, and help new contributors navigate the codebase and contribution workflow.

### Compyle (YC-backed) | Community Manager & Moderator

2025 – Present

- Manage and moderate the developer community, supporting onboarding and surfacing feedback to the engineering team.